PakCareer AI

Data-driven degree guidance for Pakistani college graduates

Deliverables 1–2: Data Foundations, Model Implementation & Evaluation (Track A)

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Abstract

Every year, students finishing intermediate or A-Levels in Pakistan choose a degree that shapes their whole career, and the choice is very often made by their parents, who default to a small set of "safe" professions. PakCareer AI sorts a student into one of seven degree families ("houses") from their interests and personality, and backs the recommendation with Pakistan-specific labour-market statistics the student can show their family. This report covers Phase 1 (problem definition) and Phase 2 (data acquisition and preprocessing). We collected five sources: 145,828 RIASEC interest-test responses (training data), the O*NET 31.0 occupation database, Pakistan Labour Force Survey indicators from ILOSTAT, World Bank indicators, and 10,999 live Pakistani job postings scraped from Rozee.pk. We also evaluated ten supplementary Kaggle career datasets and excluded all of them from training: six have random labels, one leaks the target and three are too small or unlabelled. After cleaning, the modelling table holds 81,256 labelled respondents with 30 engineered, scaled and encoded features. A preliminary gradient-boosting baseline places the student's actual field among its top three houses 86% of the time.

1 Phase 1: Problem Definition

1.1 Problem statement

In Pakistan the decision of what to study after intermediate (FSc, ICS, ICom, FA) or A-Levels is frequently taken by the family rather than the student. Medicine and engineering in particular are treated as default "respectable" careers, regardless of whether the student is interested in or suited to them. The result is students who are unmotivated in their degree, switch fields mid-way, or graduate into a career they did not want.

Choosing "any degree" is also no longer a safe bet. The Labour Force Survey data we collected (Section 5.2) shows that in 2025 the unemployment rate among Pakistanis with advanced (degree-level) education was **11.1%**, against **7.0%** for the workforce as a whole, and it peaked above 16% in 2015 and 2021. For women with a degree it was **23.9%**, compared with 6.8% for men. A degree that does not fit the student, in a field the market does not need, is costly.

Students who disagree with their parents usually have nothing but their own opinion to argue with. Career counselling is rare in Pakistani colleges, and generic online personality quizzes give no evidence a parent would take seriously.

PakCareer AI addresses both halves of the problem:

1. **Fit:** the student completes a short assessment of interests (the 48-item RIASEC inventory) and personality (TIPI). A trained classifier estimates how well they fit each of seven degree families: Medicine & Health, Engineering, Computer Science & IT, Business & Commerce, Law, Arts, Humanities & Social Sciences, and Pure Sciences.
2. **Evidence:** the recommendation comes with a statistical report built from Pakistani data, covering earnings by sector, unemployment by education level, and current job demand per field and city. The student can use it to make an informed case to their parents.

1.2 Target audience

- **Primary users:** Pakistani students aged roughly 16–20 who are finishing intermediate or A-Levels, or are in the first year of a degree and considering a switch.
- **Secondary users:** their parents, who read the statistical report, and teachers or college counsellors who want a structured, evidence-based starting point for a career conversation.

1.3 AI justification

A rule-based system would encode Holland's theory directly, e.g. "highest score Realistic ⇒ Engineering". Our data shows why that is not enough:

- **High-dimensional, overlapping inputs.** Each student is described by 58 item responses (or 30 engineered features). Houses overlap heavily. Engineering and CS & IT are *both* highest on Realistic, and Medicine and Pure Sciences are *both* highest on Investigative (Figure 5). They differ only in the combination of several weaker signals (Social, Conventional, Agreeableness, …). Hand-writing thresholds for every combination of 30 features is infeasible and would be arbitrary.
- **Noisy, probabilistic relationship.** People with the same profile end up in different fields. A useful system must output a *probability* of fit for every house, not a single hard rule, so the student and parents can see second and third choices. A learned model calibrated on 81k labelled examples gives this directly.
- **Learned weights instead of guessed ones.** How much should vocabulary, openness or age count relative to interest scores? A model estimates these weights from data and can be evaluated (accuracy, macro-F1) on held-out students. A rule system cannot be validated the same way.
- **Adaptable to Pakistan.** The training data is international (53% US). With an ML pipeline, the Pakistani survey responses we are collecting can be used to re-weight or fine-tune the model and to measure how well it transfers. A fixed rule set would have to be rewritten by hand.
- **Bias can be measured.** Because predictions are learned, we can audit them, for example by checking whether the model under-recommends Engineering to women (Figure 7a), and correct with re-weighting.

1.4 Proposed system

1. The student takes the assessment in a web app (RIASEC items, TIPI, a few background questions).
2. The same preprocessing pipeline fitted in this deliverable (`preprocessor.joblib`) transforms the answers. A classifier (Phase 3: logistic regression, random forest, gradient boosting and an MLP will be compared) outputs a probability for each house.
3. The report generator combines the top houses with the cleaned Pakistan statistics into a downloadable report for parents.

2 Phase 2: Data Acquisition

2.1 Sources

All raw data was collected by one reproducible script, `data_scripts/collect_data.py` (Python standard library only). It writes `data/raw/collection_log.json`, recording the URL, file, row count, size and timestamp of every download as evidence of collection. Table 1 summarises the sources.

Table 1: Data sources collected on 18 October 2026.

Source	Method	Size	Role
Open Psychometrics RIASEC test	ZIP download	145,828 × 94	**Training data**: 48 interest items, TIPI, vocabulary check, demographics, free-text university major (target)
O*NET 31.0 (US Dept. of Labor)	ZIP download	21 tables, ∼1,016 occupations	RIASEC profile, education and skills of each occupation; used to design per-house tasks
ILOSTAT, Pakistan LFS	SDMX REST API	9 indicators, 36,954 rows	Earnings by sector/occupation/education, unemployment by education, employment, all by sex
World Bank	REST API	16 indicators, 1990–2025	Enrolment, unemployment, inflation, exchange rate
Rozee.pk job postings	Web scraping	10,999 postings	Current Pakistani job demand per field and city; degree and experience asked for
Pakistani student survey	Google Form (ours)	ongoing	Local validation data with the same RIASEC/TIPI items plus parental-pressure questions
ESCO (European Commission)	download	3,039 occupations	Occupation taxonomy with ISCO codes (links job titles to ILO occupation groups)
Career knowledge base, Q&A and r/careerguidance sets	download	∼17k rows	Text for the planned chatbot and for explanations in the parents' report (not training data for the classifier)
Ten Kaggle career-prediction sets	download	180–21,273 rows each	Evaluated as extra training data and **excluded** (Section 2.3)

2.2 Collection details

- **APIs.** ILOSTAT data was queried through its SDMX endpoint (`sdmx.ilo.org/rest/data/ILO,<dataflow>/PAK...`) with labels included. The World Bank data comes from `api.worldbank.org/v2/country/PAK/indicator/<id>`.
- **Scraping.** Rozee.pk search pages embed the job list as JSON. The scraper walked all 550 result pages (20 jobs each), only on paths allowed by the site's `robots.txt`, with a 1.5 s delay between requests. It stored the full JSON of every posting and a flattened CSV. Scraped postings are kept out of the public repository and used only for aggregate statistics.
- **Survey.** We designed a Google Form (`survey/pakistan_student_survey.pdf`) that repeats the 48 RIASEC and 10 TIPI items word for word, so responses can be merged with the training data. It adds questions on the field the student wants, the field their parents want, who decides, and perceived family pressure. It collects no names or emails. The form is live, but no responses had arrived at the time of writing, so it is not yet part of the analysis.
- **Storage.** Raw data is too large for GitHub and is shared through Google Drive (link in the README). `.gitignore` excludes `data/`.

2.3 Supplementary datasets evaluated and excluded

To enlarge the training data we collected ten Kaggle career-prediction datasets. Each maps student attributes (skill ratings, subject marks, interests) to a recommended job. Before merging any of them, we tested whether their labels can be learned at all (`evaluate_supplementary.py`). A gradient-boosting classifier with 3-fold cross-validation predicted each dataset's own target from its own features, and a usable dataset must beat both random guessing and always predicting the most common class.

Table 2: Learnability test of the supplementary datasets (3-fold CV accuracy).

Dataset	Rows	Classes	Random	Majority	**Model**
`roo_data.csv`	20,000	34	2.9%	6.1%	**4.0%**
`dataSet.csv`	20,000	7	14.3%	28.8%	**23.6%**
`Career-prediction-data.csv`	6,901	12	8.3%	9.1%	**8.4%**
`career_path_in_all_field.csv`	9,000	90	1.1%	1.5%	**0.9%**
`career_dataset_large.xlsx`	5,000	12	8.3%	8.6%	**8.1%**
`output.csv`	1,000	10	10.0%	11.1%	**9.9%**
`Course_Job_Recommendation.csv`	21,273	45	2.2%	9.0%	**100%**
`cs_students.csv`	180	33	3.0%	10.6%	**78.3%**
`AI-based Career Rec. System.csv`	200	32	3.1%	10.5%	**25.4%**
`career_compute_dataset.csv`	7,525	–	–	–	49 labels

- **Random labels.** Six datasets score at chance level, which shows their labels were generated independently of the features (synthetic data). Training on them teaches a model noise. This explains an earlier experiment of ours that combined their categorical features and reached only 36.5% accuracy, with 0% recall for Law and Pure Sciences.
- **Target leakage.** `Course_Job_Recommendation` scores 100% because columns such as the job's role category and functional area are derived from the job title it is meant to predict. It also uses the completed degree as an input, the reverse of our task.
- **Too small or unlabelled.** `cs_students.csv` has real signal, but only 180 rows, almost all CS careers. A second copy of the same file was found and deleted. `career_compute` labels only 49 of its 7,525 rows.

All ten are kept in the shared data folder under `raw/excluded/` so the decision can be re-checked. A student-performance dataset (exam scores) was also excluded as out of scope.

3 Phase 2: Cleaning

3.1 RIASEC training data (`clean_riasec.py`)

The raw file has 145,828 rows. Figure 2a shows the missing values after recoding, and Figure 3 the two main outlier problems.

Missing values. The dataset uses `0` for "not answered" in every Likert and demographic column, even though the codebook does not define it. We recoded 0 to missing.

- Up to 4 of the 48 interest items may be missing. Each gap is filled with the person's median on the *same* scale, which preserves their profile (11,995 cells imputed).
- Respondents missing more than 4 items are dropped.
- Missing TIPI items get the column median.
- Missing categorical demographics (urban, native language, education) become an explicit "unknown" level instead of being guessed.

Invalid responses.

- **Over-claimers:** the vocabulary checklist contains three non-words. We drop anyone who claimed to know two or more of them. One is tolerated, because 20% of respondents ticked at least one, and is kept as a feature (`fake_word_flag`).
- **Speeders:** people who answered all 48 items in under 60 seconds are dropped.
- **Straight-liners:** people who gave the same answer to every item are dropped.

Outliers.

- **Age:** contains typed garbage up to 2,147,483,647. Ages outside 13–80 are dropped.

- **Family size:** contains values up to 999,999,999. Values above 20 are set to missing and median-imputed.
- **Time on test:** has a long tail (tabs left open for days). It is capped at the 99th percentile (≈66 minutes).

Duplicates. Repeated submissions with identical answers, age, gender, country and major are removed.

Target label. The free-text `major` field contains 11,816 distinct spellings (Figure 1a: "psychology", "psycology", "pyschology", "CS", …). A keyword rule set (`house_mapping.py`) maps each answer onto one of the seven houses. Rules are ordered from specific to general, so that, for example, "computer engineering" maps to CS & IT before the generic "engineer" rule fires, and "political science" maps to Social Sciences before the generic "science" rule. Non-answers ("no", "undecided", "idk") are dropped. The rules label 92.5% of all non-blank majors.

Table 3 and Figure 2b show the effect of each step.

Table 3: Rows removed at each cleaning step.

Step	Removed	Remaining
Raw download	–	145,828
Claimed ≥2 fake words	7,322	138,506
Speeders (<60 s)	272	138,234
Straight-liners	288	137,946
Age outside 13–80	74	137,872
>4 interest items skipped	212	137,660
Duplicate submissions	2	137,658
No usable major (blank or non-answer)	56,402	**81,256**

3.2 Pakistan data (`clean_pakistan.py`)

- **ILOSTAT:** the nine indicator files are combined into one long table. Codes and labels are split apart, annual and quarterly values are separated, and 4,985 empty or duplicate observations are removed (31,969 rows remain). Average earnings by ISIC sector are then mapped to houses as a proxy, e.g. Q (health) → Medicine & Health and J (information & communication) → CS & IT.
- **World Bank:** the long table is pivoted to year × indicator and empty years are dropped.
- **Rozee.pk:**
  - 1,405 postings repeated with the same title, company and city are removed. They are mostly re-posts aggregated from other boards. 9,580 postings remain.
  - Experience text is parsed into years ("Fresh" becomes 0).
  - The minimum degree is found in only 1,833 structured fields, so it is also extracted from the description text with regular expressions, giving a level for 7,589 postings.
  - Job titles are mapped to houses with the same rule set. 7,469 postings are mapped; the rest are mostly non-degree jobs such as drivers and guards.

4 Phase 2: Feature Strategy (`build_features.py`)

4.1 Engineered features

Table 4: Model inputs (30 columns after encoding).

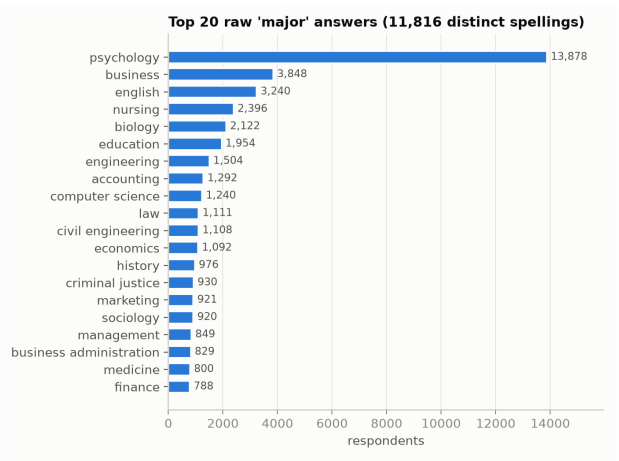
Feature	Construction	Treatment
RIASEC scores (6)	Mean of the 8 items of each scale (1–5)	StandardScaler
Big Five (5)	TIPI pairs, second item reverse-scored: $(x_a + (8 - x_b))/2$	StandardScaler
`vocab_score`	Number of the 13 real words known	StandardScaler
`differentiation`	Highest minus lowest RIASEC score (how clear-cut the profile is)	StandardScaler
`familysize`	Children of the respondent's mother	StandardScaler
`age`	Right-skewed (median 21)	$\log(1 + x)$, then StandardScaler
`education`	$1 < 2 < 3 < 4$, unknown $= 0$	Ordinal encoding, scaled
`holland_code`	Letter of the highest RIASEC score	One-hot (6)
`urban`, `engnat`	Childhood area, native English speaker	One-hot incl. "unknown"
`fake_word_flag`	Claimed exactly one non-word	Binary, passthrough
`house` (target)	7 classes	Label encoding (0–6)

4.2 Scaling, encoding and leakage

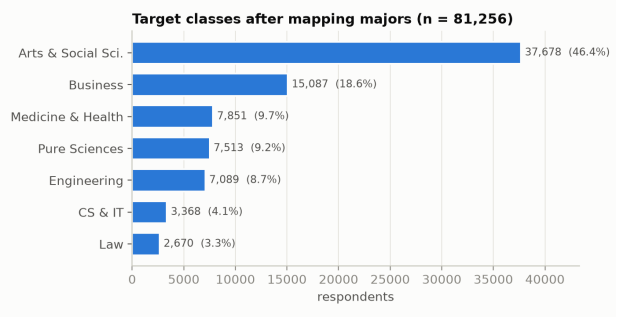
- **Pipeline.** All transformations are one scikit-learn `ColumnTransformer`. Z-score standardisation was chosen over min–max scaling because the scale scores are roughly bell-shaped (Figure 4) and the planned linear and neural models benefit from zero-mean inputs.
- **Split and leakage.** The data is split 80/20, stratified by house (65,004 training, 16,252 test). The scalers and encoders are fitted on the training split *only*, then applied to the test split, so no test-set statistics leak into training. The fitted pipeline is saved for use in the web app.
- **Class imbalance.** Arts, Humanities & Social Sciences holds 46.4% of respondents and Law 3.3%, a 14:1 ratio (Figure 1b). We pre-computed balanced class weights (0.31 for Arts up to 4.35 for Law) for Phase 3, and will report macro-F1 rather than accuracy alone.
- **Ethics.** Gender, religion, race, sexual orientation and country are *not* model inputs. Gender and a Pakistan flag are stored beside the split only for the Phase 4 bias check.

5 Exploratory Data Analysis

5.1 Training data

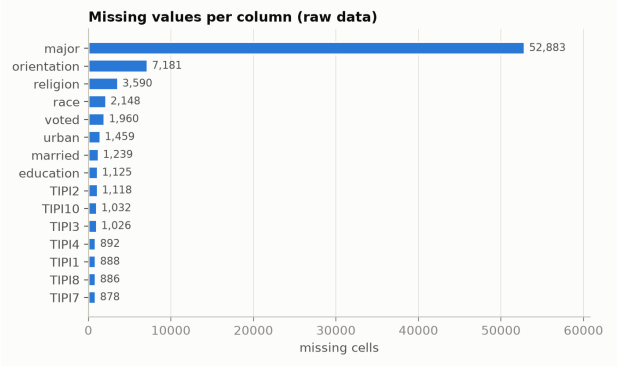


(a) Raw free-text majors before mapping.

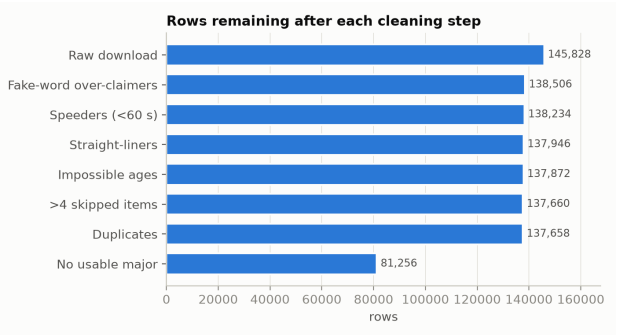


(b) Classes after mapping: 14:1 imbalance.

Figure 1: From 11,816 messy spellings to seven target classes.



(a) Missing values per column (raw).



(b) Rows remaining after each step.

Figure 2: Data quality before cleaning. Most missing majors are blank answers. Orientation, religion and race are missing more often but are not used.

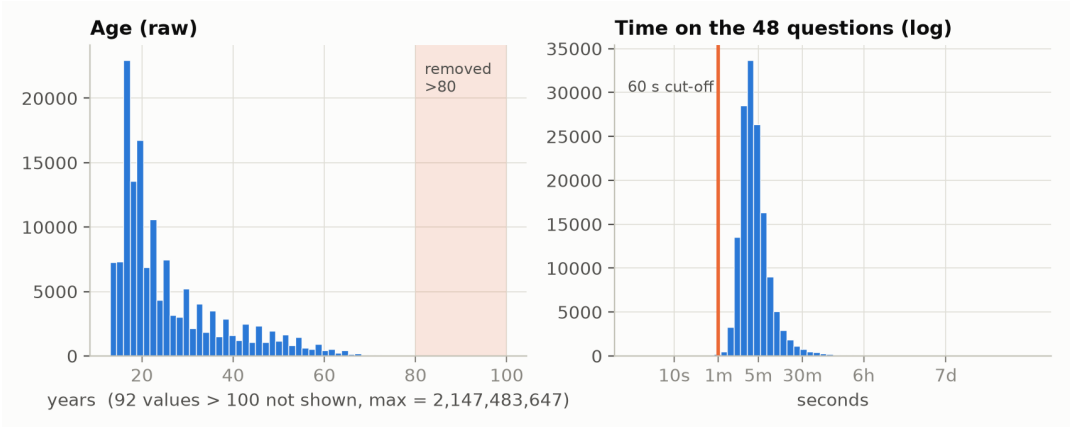


Figure 3: Outliers. Left: ages above 80 (and 92 values above 100, up to $2^{31} - 1$) are removed. Right: time spent on the test. Answers under 60 s (orange line) are treated as speeders.

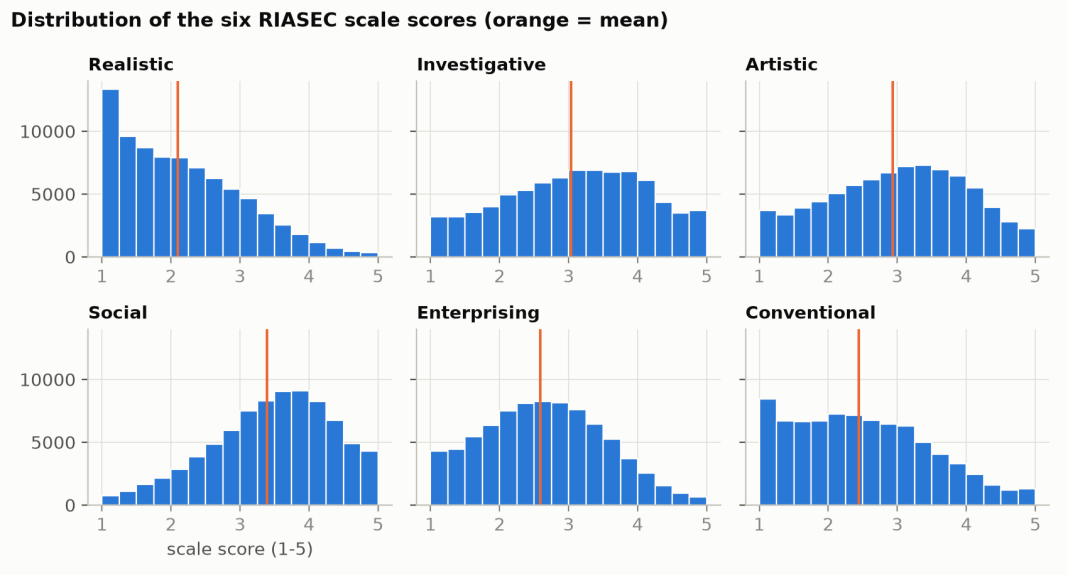


Figure 4: The six RIASEC scale scores. Social and Investigative lean high and Realistic and Conventional lean low. This reflects the sample (two thirds women, many psychology majors), not the population, which is one more reason for class weighting.

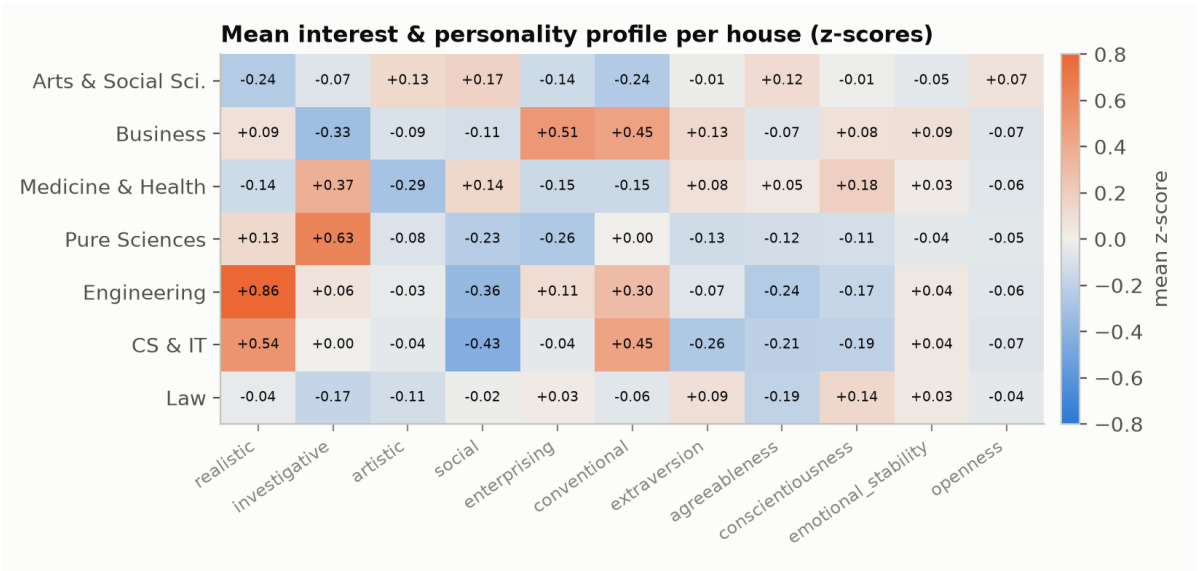


Figure 5: Mean profile of each house in z-scores. The data reproduces Holland's theory. Engineering and CS are highest on Realistic, Medicine and Pure Sciences on Investigative, Business and Law on Enterprising, and Arts & Social Sciences on Social. The overlaps between pairs of houses are why a single-rule system is not enough.

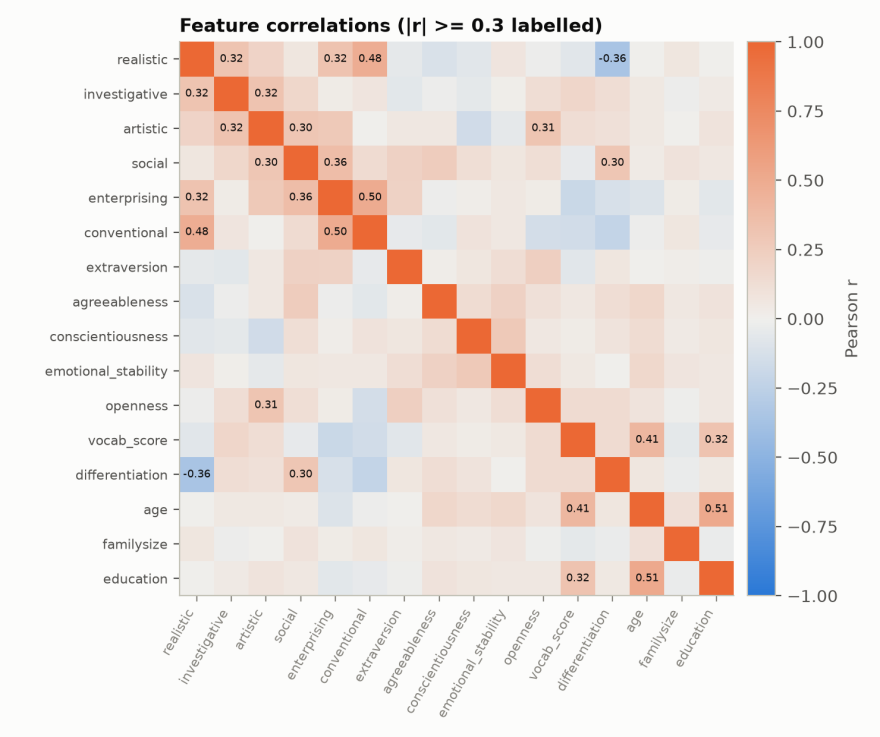
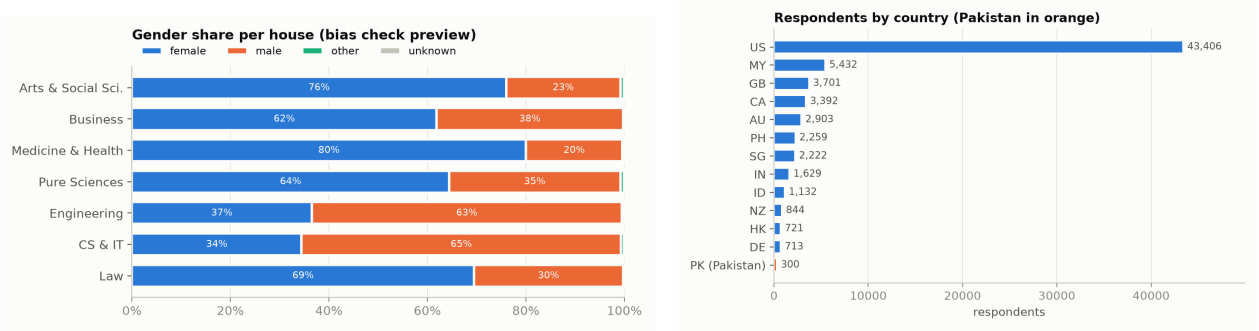


Figure 6: Feature correlations. The strongest pairs are age–education ($r = 0.51$), Enterprising–Conventional (0.50) and Realistic–Conventional (0.48), matching the adjacency of these types on Holland's hexagon. No pair is high enough ($|r| > 0.8$) to require removing a feature.



(a) Gender share per house.

(b) Country of respondents.

Figure 7: Two sources of bias to address in Phase 4. (a) Engineering and CS are about two-thirds male, so a model could learn to recommend them less to women. Gender is therefore excluded from the inputs and per-gender metrics will be reported. (b) 53% of respondents are from the US and only 300 are from Pakistan, a domain shift our own survey is designed to measure.

5.2 Pakistan statistics

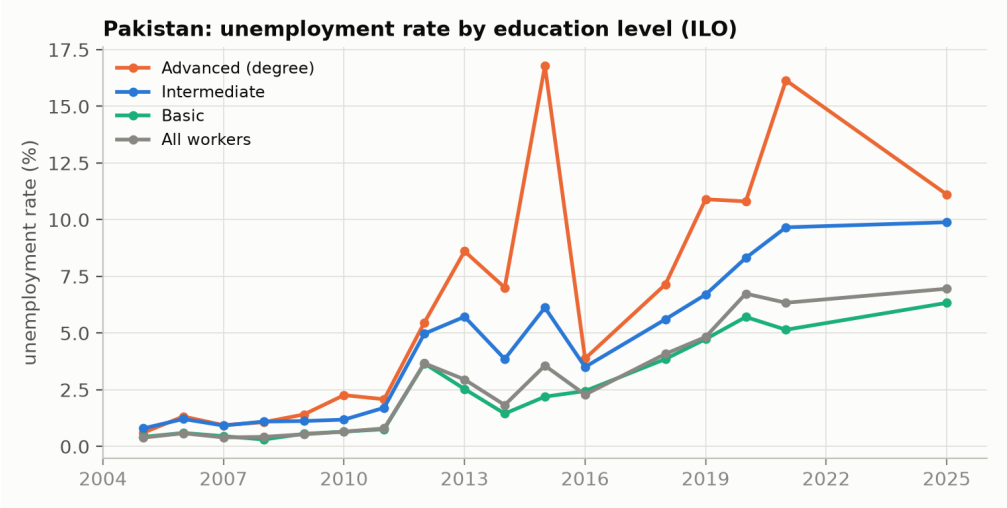
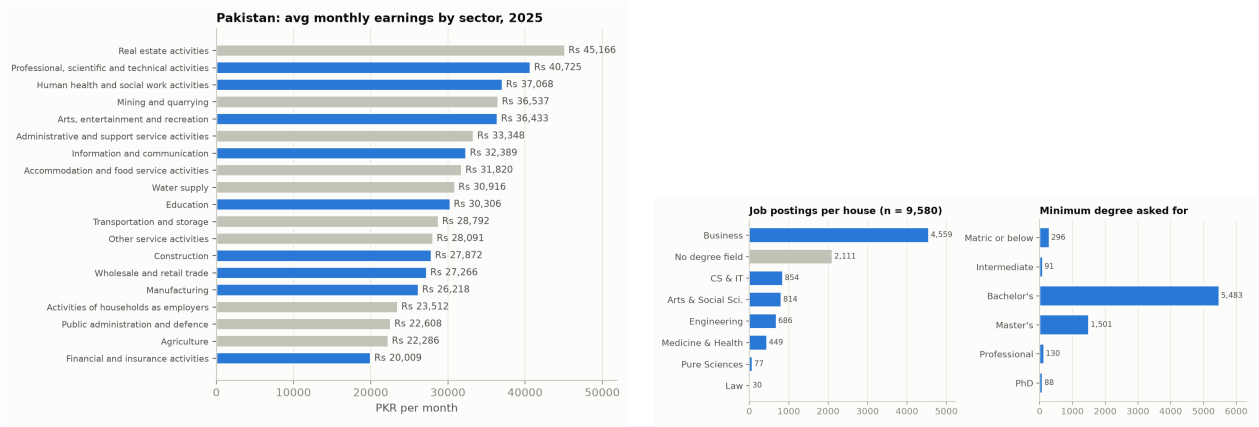


Figure 8: Unemployment rate by education level in Pakistan (ILOSTAT, Labour Force Survey). Since 2012, people with a degree have consistently had *higher* unemployment than the workforce as a whole, reaching 11.1% in 2025. The jumps in 2015 and 2016 coincide with changes in survey rounds and should be read with care.



(a) Average monthly earnings by sector, 2025.

(b) Rozee.pk postings by house and degree asked for.

Figure 9: (a) Sector averages include all employees in a sector (a hospital's cleaners as well as its doctors), so they are a coarse proxy for a degree's value. The low figure for finance in particular needs checking against occupation-level data in Phase 3. (b) Business roles dominate current postings (sales and marketing), and 72% of postings that state a degree ask for a bachelor's.

6 Preliminary Baseline (preview of Phase 3)

To check that the prepared data supports the task, we trained quick baselines on the split from Section 4 (`baseline_models.py`). Because the app will show every student a ranked list of houses, we report top-k accuracy (is the true house among the k most probable?) next to ordinary accuracy and macro-F1.

Table 5: Baselines on the held-out test set (16,252 students, 7 houses).

Model	Top-1	Top-2	Top-3	Macro-F1
Always predict the largest house	46.4%	–	–	–
Logistic regression, class-balanced	36.8%	55.1%	71.4%	0.30
Gradient boosting, 30 features	53.0%	73.0%	83.4%	0.28
Gradient boosting, 30 features + 58 raw answers	**56.6%**	**76.2%**	**86.0%**	**0.35**

- **Top-1 accuracy alone is misleading.** Always predicting Arts & Social Sciences already scores 46.4%. Class balancing lowers top-1 accuracy (36.8%) but raises macro-F1, because the model stops ignoring Law and CS.
- **Interests predict a field only moderately.** People with the same interest profile study different subjects, so a single-answer accuracy of 70–80% is not realistic from this data. Restricting training to graduates did not change the result. The best model's top-3 list contains the student's actual field **86%** of the time, which supports presenting the top three houses with probabilities rather than a single verdict.
- **Where further gains must come from:** new signals rather than more of the same data. These are the task-based career simulations planned for the app, the Pakistani survey, and tuning of the gradient-boosting and neural models in Phase 3.

7 Phase 3: Model Architecture & Implementation

7.1 Framework Selection

We implement the classifier using **XGBoost** (v3) and **scikit-learn** (v1.9), both industry-standard frameworks for supervised learning on tabular data. A neural-network approach (PyTorch MLP) was considered but rejected: our feature set is fully engineered and structured, the dataset has no spatial or sequential structure, and gradient-boosted trees are empirically superior on tabular classification tasks of this scale [8]. XGBoost was preferred over sklearn's `GradientBoostingClassifier` because its histogram-based split finding reduces training time by an order of magnitude on 65 000 rows (the final model trains in under a minute).

7.2 Architecture Definition

The final model is an **XGBoost gradient-boosted tree ensemble**. Gradient boosting builds an additive model

$$F_M(x) = \sum_{m=1}^{M} \eta \, h_m(x),$$

where each weak learner h_m is a regression tree fitted to the pseudo-residuals of the previous iteration, η is the learning rate, and M is the number of boosting rounds. For multi-class classification with $K = 7$ houses the loss is the **softmax cross-entropy**

$$\mathcal{L} = -\sum_{i=1}^{N} \sum_{k=1}^{K} y_{ik} \log \hat{p}_{ik},$$

and predictions are the argmax of the K output probability vectors. Figure 10 shows the end-to-end pipeline.

Class imbalance. The classes are imbalanced 14:1 (Arts vs. Law). We trained two variants: an *unweighted* model and a *class-balanced* model whose samples are weighted with `compute_sample_weight('balanc`
Balancing raises macro-F1 and minority-class recall but lowers top-k accuracy (Table 7). Because

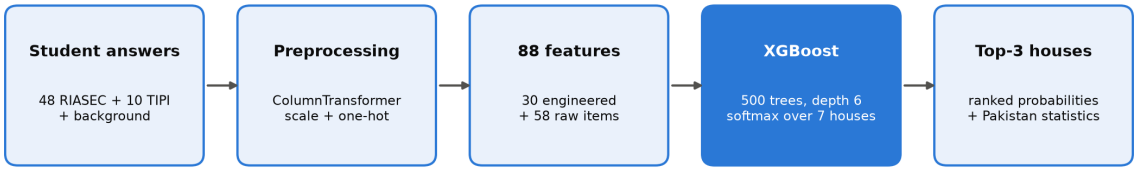


Figure 10: End-to-end pipeline: a student's answers are preprocessed by the fitted scikit-learn `ColumnTransformer`, fed into the XGBoost ensemble, and the softmax probabilities are ranked to produce the top-3 house recommendations.

the application presents a ranked top-3 list, we select the **unweighted** model as the final model and report the balanced variant as a comparison.

7.3 Feature Set

The model is trained on **88 features** derived from two groups:

- **30 engineered features**: six RIASEC domain scores, five Big Five (TIPI) trait scores, vocabulary score, profile differentiation, Holland code one-hot encoding, and demographic variables (age, education, family size, urban/rural, English nativeness).
- **58 raw item responses**: 48 RIASEC Likert-scale answers (R1–C8) and 10 TIPI personality items (TIPI1–TIPI10), retained because they provide the per-item signal that domain averages suppress.

Sensitive attributes (gender, religion, race, sexual orientation, country and the Pakistan flag `is_pakistan`) are *excluded* from model inputs by `split_features` in `train.py` and are retained only for the bias audit (§8.7). An earlier training run had included `is_pakistan` by mistake; removing it changed no metric by more than 0.1 percentage points.

7.4 Hyperparameter Tuning

A `RandomizedSearchCV` with 5 configurations and 3-fold stratified cross-validation was run first to identify promising regions of the search space; the winning configuration was then fixed and retrained on the full training split. Table 6 documents the values explored and the final selection.

Table 6: Hyperparameter search space and final values.

Parameter	Values Tested	Final	Justification
`n_estimators`	50, 200, 300, 400, 500	500	Validation loss still falling past round 300
`max_depth`	3, 5, 6, 7	6	Depth 3 underfits; depth 7 overfits minority classes
`learning_rate`	0.01, 0.05, 0.1	0.05	Lower rate with more trees beats high rate with few
`subsample`	0.7, 0.8, 1.0	0.8	Row subsampling reduces variance
`colsample_bytree`	0.6, 0.8, 1.0	0.8	Feature subsampling decorrelates trees
`min_child_weight`	5, 10, 20	10	Prevents splits on tiny Law leaf nodes

8 Phase 4: Training & Evaluation

8.1 Validation Strategy

The dataset was partitioned into training (65 004 rows) and test (16 252 rows) splits using stratified sampling in Phase 2, ensuring each of the seven houses is represented proportionally in both splits. During training, 15% of the training split is held out as a validation set for the loss curves; during

hyperparameter search we applied **stratified cross-validation** on the training split. Final performance figures are reported on the held-out test split, which the model never saw during training or search (`src/evaluation/evaluate.py`).

8.2 Metric Analysis

We report five metrics and justify each for this problem.

- **Top-1 / Top-2 / Top-3 accuracy.** The application presents students with their top-3 recommended houses. Top-k accuracy directly measures the product promise: if the correct house appears in the top-3, the student receives actionable guidance.
- **Macro-F1.** With a 14:1 class imbalance, accuracy is dominated by the majority class (Arts & Social Sciences). Macro-F1 averages the per-class F1-score with equal weight per class, penalising the model for failing on minority classes such as Law (534 test samples).
- **AUC-ROC (macro one-vs-rest).** Measures how well the model's probability scores separate each house from the remaining six, which matters because the app displays probabilities to students and parents.

8.3 Results

Table 7 compares the Phase 2 baselines against the final XGBoost model. The final model exceeds every baseline on top-k accuracy and macro-F1.

Table 7: Baselines vs. final model on the held-out test set (16 252 students, 7 houses).

Model	Top-1	Top-2	Top-3	Macro-F1
Always predict majority class	46.4%	—	—	—
Logistic Regression (Phase 2)	36.8%	55.1%	71.4%	0.30
Gradient Boosting, 30 feat. (Phase 2)	53.0%	73.0%	83.4%	0.28
Gradient Boosting, 88 feat. (Phase 2)	56.6%	76.2%	86.0%	0.35
XGBoost 88 feat., class-balanced	47.9%	68.5%	81.1%	**0.382**
XGBoost 88 feat. (final)	**57.1%**	**76.5%**	**86.4%**	0.358
AUC-ROC, final model (macro OvR)		0.798		

The confusion matrix is shown in Figure 11.

8.4 Training Curves

Figure 12 shows the softmax cross-entropy log-loss on the training and validation splits across all 500 boosting rounds.

8.5 Cross-Validation Results

To confirm the test-set results are not an artefact of a single split, we ran 5-fold stratified cross-validation on the full training set (65 004 rows) using the final hyperparameter configuration with `n_estimators=200` to keep runtime tractable.

Table 8: 5-fold stratified cross-validation on the training split.

Metric	Mean	Std Dev
Macro-F1	0.343	$\pm$ 0.005
Accuracy	0.561	$\pm$ 0.003

The low standard deviation ($\pm$0.005 on macro-F1 across five folds) confirms that performance is stable and not sensitive to the choice of split. The CV macro-F1 of 0.343 (200 trees) is consistent with the held-out test macro-F1 of 0.358 (500 trees), confirming no overfitting to the test split.

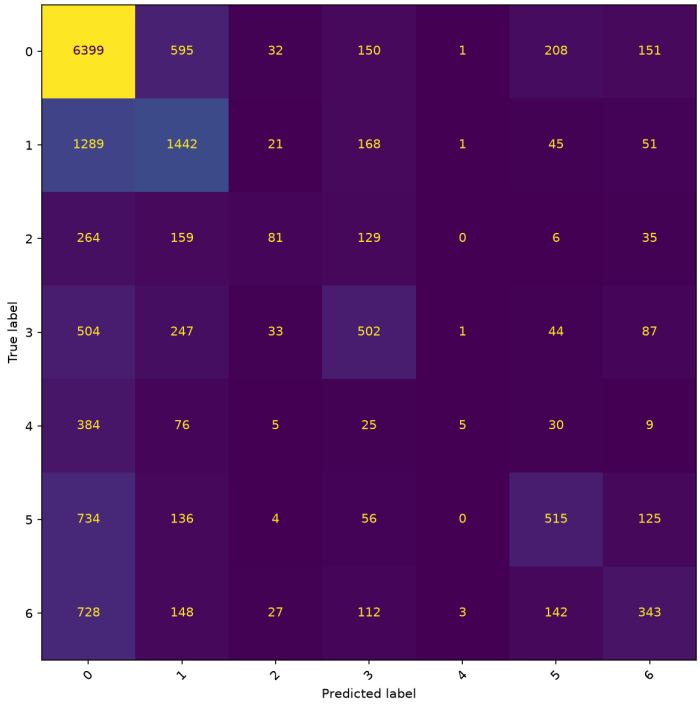


Figure 11: Confusion matrix of the final model on the 16 252-student test set. Rows are true houses; columns are predicted houses. Class indices: 0 = Arts, Humanities & Social Sciences, 1 = Business & Commerce, 2 = CS & IT, 3 = Engineering, 4 = Law, 5 = Medicine & Health, 6 = Pure Sciences.

8.6 Error Analysis

Per-class recall from the confusion matrix shows where the final model struggles (Table 9).

Table 9: Per-class recall of the final model on the test set.

House	Test samples	Recall	Most often predicted as
Arts, Humanities & Social Sciences	7 536	84.9%	Business (595)
Business & Commerce	3 017	47.8%	Arts (1 289)
Engineering	1 418	35.4%	Arts (504)
Medicine & Health	1 570	32.8%	Arts (734)
Pure Sciences	1 503	22.8%	Arts (728)
CS & IT	674	12.0%	Arts (264)
Law	**534**	**0.9%**	Arts (384)

Three patterns emerge:

1. **The majority class absorbs uncertain students.** Arts & Social Sciences is 46.4% of the data, and the model predicts it for 10 302 of 16 252 test students. Every other house loses most of its errors to Arts.
2. **Law is almost never predicted (recall 0.9%).** Only 11 test students are predicted as Law. Law students' RIASEC profiles overlap heavily with Social and Enterprising types that dominate Arts and Business, and Law is the smallest class. The class-balanced variant raises Law recall to 17.6% and CS & IT recall from 12.0% to 35.3%, at the cost of 9 points of top-1 accuracy.
3. **Top-3 is far more reliable than top-1.** Because the true house is usually the model's second or third choice when it is not the first, the top-3 list contains it 86.4% of the time.

These failure modes motivate two planned improvements: (i) the task-based career simulation features described in the product roadmap, which should provide house-discriminative behavioural signal beyond self-report scores, and (ii) the Pakistani student survey data, which will anchor the

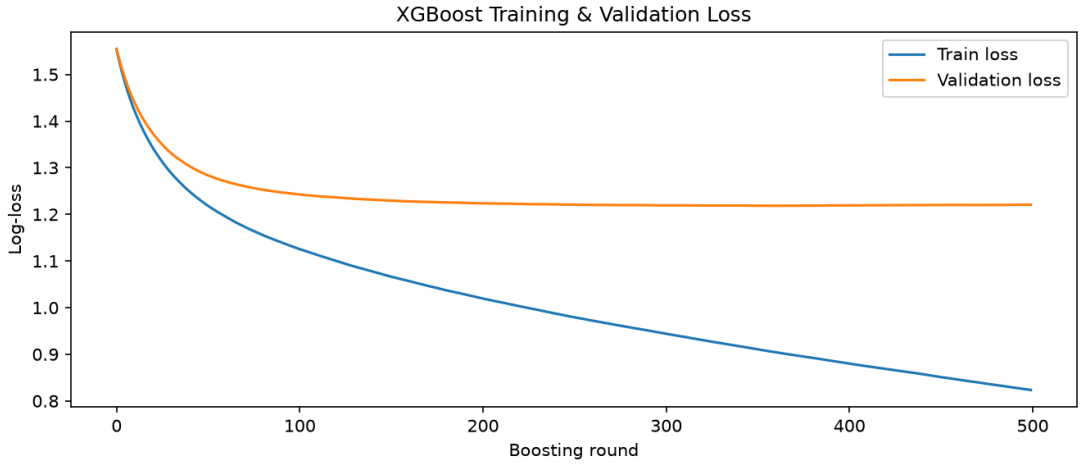


Figure 12: XGBoost log-loss per boosting round. Validation loss reaches its minimum around round 370 (1.218) and stays flat afterwards (1.221 at round 500), while training loss keeps decreasing. The model has reached its generalisation limit; early stopping near round 370 would save about 25% of training time without changing test performance.

model to local degree norms.

8.7 Bias Check

We audited model performance across two sensitive attributes: **gender** and **Pakistan-residency status** (`is_pakistan`) with `src/utils/bias_check.py`. Neither attribute is an input to the model; they are used solely for this post-hoc analysis. Figure 13 and Table 10 summarise the results.

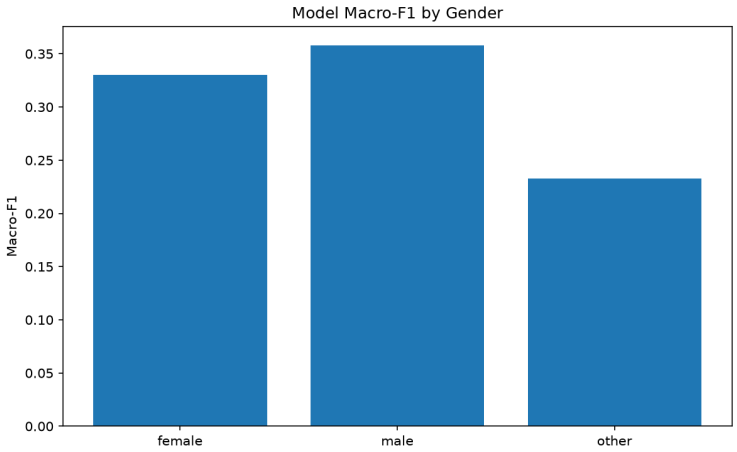


Figure 13: Macro-F1 by gender subgroup.

Gender. The model performs somewhat better for male (0.358) than female (0.330) students, a gap of 0.028. A likely cause is the gender skew within houses (Figure 7a): Engineering and CS are about two-thirds male, and these are the houses the model already finds hardest, so errors there weigh differently on each group. The "other" gender group scores lower (0.233), but its sample is too small ($n = 76$) for firm conclusions.

Pakistan residency. Pakistani respondents ($n = 62$) show a macro-F1 of 0.271 compared to 0.359 for non-Pakistani respondents. Two factors explain this: (i) the training data comes predominantly from a Western, English-speaking population (Open Psychometrics), so the model has seen few

Table 10: Macro-F1 by sensitive subgroup (final model).

Attribute	Group	n	Macro-F1
Gender	Female	10 978	0.330
Gender	Male	5 179	0.358
Gender	Other	76	0.233
Pakistan residency	Non-Pakistan	16 190	0.359
Pakistan residency	Pakistan	62	0.271

Pakistani interest profiles, and (ii) 62 students is too small a sample for a reliable estimate. The Pakistani student survey (Section 2) is designed to close this gap in future training iterations.

Mitigation already in place. Sensitive attributes are excluded from model inputs, preventing direct discrimination, and the class-balanced variant is available if minority-house recall is prioritised. The bias audit will be repeated once the Pakistani survey data is incorporated, and reweighting by region and gender will be applied if the gaps persist.

9 Team Contribution

Table 11 maps each Git commit author to the modules they committed, taken from the repository history.

10 Challenges and Limitations

- **Label noise.** Majors are self-reported free text, and some respondents are students who may still switch. The keyword mapping is transparent and testable, but imperfect.
- **Domain shift.** The training data is mostly from the US and the users are Pakistani. Only 300 Pakistani rows exist. Our survey is the planned fix, but it has no responses yet.
- **Proxy statistics.** No public dataset gives earnings per *degree* in Pakistan, so sector-level ILO data and job postings stand in for it. PBS microdata, which must be requested, would allow degree-level figures.
- **Salaries on Rozee.pk** are hidden behind a login for most postings, so salary comes only from ILO data.

11 Next Steps

Deploy the final XGBoost model behind an API with a hosted web front end that shows each student their top-3 houses with probabilities and the Pakistan statistics report. Add the task-based career simulations as new features, incorporate the Pakistani survey responses, and repeat the bias check.

Table 11: Git commit author to project module mapping.

Author	Module / File	Description
Muhammad Hasan Butt (24L-3022)	`data_scripts/clean_pakistan.py`	ILO, World Bank and Rozee.pk cleaning
	`data_scripts/build_features.py`	Feature engineering, scaling; added 58 raw item columns
	`data_scripts/evaluate_supplementary.py`	Supplementary-dataset test, Phase 2 baselines
	`baseline_models.py`	
	`src/models/model.py`, `src/models/train.py`	Model architectures, XGBoost training, loss curves
	`report/phase3_4_sections.tex`	Phase 3–4 report sections
Kabeer Ahmed Shahzeb (24L-3087)	`README.md`, `requirements.txt`, `.gitignore`	Project setup
	`data_scripts/collect_data.py`	Data collection (APIs, scraping), collection log
	`data_scripts/house_mapping.py`, `clean_riasec.py`	RIASEC cleaning, major-to-house mapping
	`src/evaluation/evaluate.py`	Top-k accuracy, AUC-ROC, confusion matrix
	`src/models/train.py` (fix)	Restored shared helpers; excluded sensitive columns
	`report/main.tex`	Report integration and this table
Muhammad Moosa Khan Rokhri (24L-3070)	`survey/`	Pakistani student survey specification
	`data_scripts/eda.py`, `report/figures/`	Exploratory data analysis and figures
	`report/main.tex`, `main.pdf` (Deliverable 1)	Deliverable 1 report
	`src/utils/bias_check.py`*	Gender and Pakistan-status subgroup audit

*Moosa's assigned module; committed by Kabeer on his behalf at the deadline.

Repository

`data_scripts/collect_data.py`	downloads all raw sources, writes the collection log
`data_scripts/house_mapping.py`	free text → house rules
`data_scripts/clean_riasec.py`	missing values, validity, outliers, duplicates, labels
`data_scripts/clean_pakistan.py`	ILO, World Bank and Rozee.pk cleaning
`data_scripts/build_features.py`	feature engineering, scaling, encoding, split
`data_scripts/eda.py`	all figures in this report
`data_scripts/evaluate_supplementary.py`	learnability test of the Kaggle datasets (Table 2)
`data_scripts/baseline_models.py`	preliminary baselines (Table 5)

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